DO NOW — RECALL

Synoptic programming challenge

Name: _____ Class: _____ Date: _____

1. Give the Big O of a full scan of n values.

2. Which item does a queue remove first?

3. What does return count do?

4. Starting total=0, sum even values from [3,4,6].

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

O(n).

2. Model answer

The oldest enqueued item.

3. Model answer

Returns count to the caller.

4. Model answer

10 (4+6).

Vocabulary — know these terms

decomposition — Break a problem into smaller tasks.

threshold — Value used as a cutoff.

counter — Variable that records how often an event occurs.

boundary test — Input at or next to a permitted limit.

erroneous input — An input violating the program's constraints.

NEW LEARNING

Read, annotate and explain

Inputs and output

Inputs are an array and a threshold; the result is the number of qualifying items.

Check each element

The counter increases only for values greater than or equal to the threshold.

Test a boundary

A value equal to the threshold must count. For 5,8,3,8 at threshold 8, result is 2.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Count values at least 7 in [4,7,9].

Predict

Skip 4 because it is below 7.

Explain

Count 7 and 9: the result is 2.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Find CountAtLeast([2,5,7],5).

Question 2

Change the method so it counts values strictly greater than the threshold.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Give normal, boundary and erroneous test cases if the threshold must be non-negative.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The test score>=t excludes values equal to t.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.